Person-to-Project Assignment as a Mixed-Integer Linear Program: A Progressive Family of Formulations

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Abstract

We formulate the problem of assigning people to projects as a mixed-integer linear program (MILP). Rather than a single monolithic model, we develop a *progression of variants*: a static binary assignment model with hard skill-coverage constraints (Variant 1); a time- and role-indexed model with continuous hour allocations (Variant 2); and a model with soft coverage, project selection, and shortfall penalties (Variant 3). We then present cross-cutting constraint modules — pairwise compatibility, team composition and diversity, fairness, development goals, location — that can be attached to any variant, and close with a discussion of multi-objective design and computational considerations.

Contents

1 Problem Statement

An organization must staff a portfolio of projects from a pool of people. Each project demands certain skills at minimum proficiency levels, a headcount per role, and effort over a planning horizon; it carries a strategic value, a budget, and a timeline. Each person has skill proficiencies, limited capacity, a cost rate, a seniority level, preferences over projects, and a bound on concurrent engagements. The planner must decide *who works on what* (and, in richer variants, *in which role, in which period, for how many hours*) so that every project's requirements are *covered*, while trading off skill fit, preference satisfaction, cost, and strategic value.

The defining structural feature is the **coverage constraint**: a project's required skill set must be covered by the *union* of the skills of its assigned team, not by any single person. This makes the problem a generalization of set covering intertwined with a generalized assignment problem, and it is NP-hard in general.

2 Notation

2.1 Sets and indices

Symbol	Meaning
$i \in I$	people
$j \in J$	projects
$s \in S$	skills
$r \in R$	roles (e.g. lead, contributor, reviewer)
$t \in T$	time periods (weeks, sprints, months)
$d \in D$	teams / departments; $I_d \subseteq I$ is the set of people in d
$l \in L$	seniority levels (ordered)
$J^{\text{man}} \subseteq J$	mandatory (must-staff) projects
$T_j \subseteq T$	active periods of project j (from $start_j$ to end_j)

2.2 Person parameters

Symbol	Meaning
$prof_{is} \in \{0, \ldots, 5\}$	proficiency of person i in skill s
$cap_{it} \geq 0$	available hours (or FTE fraction) of i in period t
$c_i \geq 0$	cost rate of i (per hour)
$lev_i \in L$	seniority level of i
tz_i	time zone / location of i
$p_{ij} \in \mathbb{R}$	preference (interest) score of i for project j
$cert_{ik} \in \{0, 1\}$	person i holds certification/clearance k
$\overline{n}_i \in \mathbb{Z}_+$	max number of concurrent projects for i
$g_{is} \in \{0, 1\}$	i wants to *grow* skill s (development goal)

Existing commitments are folded into cap_{it} (capacity net of prior obligations). Language proficiency and domain/client experience are treated as additional "skills" in S, which keeps the model uniform.

2.3 Project parameters

Symbol	Meaning
$\pi_{js} \in \{0, \ldots, 5\}$	minimum proficiency in skill s required by j ($0 =$ not needed)
$n_{js} \in \mathbb{Z}_+$	number of assigned people who must meet skill s (coverage depth)
$\delta_{jr} \in \mathbb{Z}_+$	required headcount in role r on project j
$e_{jt} \geq 0$	effort (hours) needed by j in period t
$v_j \geq 0$	strategic value / priority of j
$B_j \geq 0$	budget cap of j
$\underline{N}_j, \overline{N}_j$	min / max team size of j
$\lambda_j \in L$	minimum seniority required for the lead role on j
$creq_{jk} \in \{0, 1\}$	project j requires certification/clearance k

2.4 Relational parameters

Symbol	Meaning
$f_{ij} \in \mathbb{R}$	skill-fit score of person i for project j (precomputed)
$P^- \subseteq I \times I$	pairs who *cannot* work together
$P^+ \subseteq I \times I$	pairs who *must* work together (if either is assigned)
$\sigma_{ii'} \in \mathbb{R}$	synergy score for pairing i and i' (e.g. prior collaboration)
$\rho_d \in (0, 1]$	max fraction of any team drawn from department d

Remark 2.1 (Derived eligibility sets). Much of the combinatorics is best handled by *preprocessing*. Define

$$Q_{js} = \{\, i \in I : prof_{is} \geq \pi_{js} \,\} \qquad \text{(people qualified for skill } s \text{ on project } j\text{)},$$

$$E_j = \{\, i \in I : cert_{ik} \geq creq_{jk} \ \forall k,\ \text{location/tz compatible with } j \,\} \qquad \text{(people eligible for project } j\text{)}.$$

Variables are only created for (i, j) with $i \in E_j$; ineligible assignments are never instantiated. This is both a correctness device and the single most effective way to keep instances small.

3 Variant 1: Static Assignment with Hard Coverage

The base model answers one question: *which people staff which projects?* Time, roles, and hours are aggregated away; each project j exposes an estimated per-person workload w_j (hours over its lifetime), so the cost of assigning i to j is $c_{ij} = c_i \, w_j$.

3.1 Decision variables

$$x_{ij} \in \{0,1\} \qquad \text{person } i \text{ is assigned to project } j \quad (i \in E_j).$$

3.2 Formulation

$$\max_{x} \quad \alpha \sum_{j\in J} \sum_{i\in E_j} f_{ij}\, x_{ij} \;+\; \beta \sum_{j\in J} \sum_{i\in E_j} p_{ij}\, x_{ij} \;-\; \gamma \sum_{j\in J} \sum_{i\in E_j} c_{ij}\, x_{ij} \tag{1}$$

$$\text{s.t.} \quad \sum_{i\in Q_{js}\cap E_j} x_{ij} \;\ge\; n_{js} \qquad \forall j\in J,\; s\in S:\; \pi_{js}\ge 1 \tag{2}$$

$$\underline{N}_j \;\le\; \sum_{i\in E_j} x_{ij} \;\le\; \overline{N}_j \qquad \forall j\in J \tag{3}$$

$$\sum_{j:\, i\in E_j} x_{ij} \;\le\; \overline{n}_i \qquad \forall i\in I \tag{4}$$

$$\sum_{i\in E_j} c_{ij}\, x_{ij} \;\le\; B_j \qquad \forall j\in J \tag{5}$$

$$x_{ij}\in\{0,1\} \qquad \forall j\in J,\; i\in E_j.$$

Constraint (2) is the **skill-coverage** constraint: for every skill s that project j requires, at least n_{js} of the assigned people must be qualified in s (the common case is $n_{js} = 1$; setting $n_{js} = 2$ buys redundancy for critical skills). Constraint (3) bounds team size, (4) caps each person's concurrent engagements, and (5) enforces project budgets.

3.3 Explicit coverage variables (skill-load control)

Formulation (2) lets a single versatile person "count" toward every skill simultaneously, which may be unrealistic: one person cannot in practice *own* five skill areas on one project. To cap the number of skill requirements a person may cover, introduce

$$y_{ijs} \in \{0,1\} \quad : \text{person } i \text{ is designated to cover skill } s \text{ on project } j \quad (i \in Q_{js}\cap E_j),$$

and replace (2) with

$$\sum_{i\in Q_{js}\cap E_j} y_{ijs} \;\ge\; n_{js} \qquad \forall j,\; s:\; \pi_{js}\ge 1 \tag{6}$$

$$y_{ijs} \;\le\; x_{ij} \qquad \forall j,\; s,\; i\in Q_{js}\cap E_j \tag{7}$$

$$\sum_{s:\, i\in Q_{js}} y_{ijs} \;\le\; \kappa \qquad \forall j,\; i\in E_j, \tag{8}$$

where κ is the maximum number of skill areas one person may own on a single project. This pattern — an explicit "who covers what" layer under the assignment layer — recurs whenever coverage must be attributed rather than merely counted.

Remark 3.1 (When Variant 1 suffices). Use Variant 1 for annual or quarterly portfolio staffing where projects are long-lived relative to the planning cycle, workloads are roughly uniform, and the interesting decision is team *composition*, not scheduling.

4 Variant 2: Time- and Role-Indexed Allocation

When projects have effort profiles over time, people have fluctuating availability, and roles matter (a project needs *a lead* and *two contributors*, not just three bodies), the assignment must be indexed by role and period, and hours become a continuous decision.

4.1 Decision variables

$$\begin{aligned}
x_{ijrt} &\in \{0,1\} && \text{person } i \text{ works on project } j \text{ in role } r \text{ during period } t \quad (i \in E_j,\ t \in T_j),\\
h_{ijt} &\geq 0 && \text{hours of person } i \text{ allocated to project } j \text{ in period } t,\\
a_{ijt} &\in \{0,1\} && \text{person } i \text{ is active on project } j \text{ in period } t \text{ (aggregation of roles)},\\
z_{ij} &\in \{0,1\} && \text{person } i \text{ is \textit{ever} assigned to project } j \text{ (membership indicator)}.
\end{aligned}$$

The auxiliary variables a_{ijt} and z_{ij} are defined by linear linking constraints below; they let concurrency, coverage, and continuity be stated cleanly.

4.2 Formulation

$$
\begin{aligned}
\text{maximize}\quad & \alpha\sum_{i,j} f_{ij}\, z_{ij} \;+\; \beta\sum_{i,j} p_{ij}\, z_{ij} \;-\; \gamma\sum_{i,j,t} c_i\, h_{ijt} \;-\; \mu\sum_{i,j} z_{ij} && && (9)\\
\text{s.t.}\quad & \sum_{r\in R} x_{ijrt} \;=\; a_{ijt} && \forall i,\ j,\ t\in T_j && (10)\\
& a_{ijt} \;\le\; z_{ij} && \forall i,\ j,\ t\in T_j && (11)\\
& \sum_{i\in E_j} x_{ijrt} \;\ge\; \delta_{jr} && \forall j,\ r,\ t\in T_j && (12)\\
& \sum_{i\in Q_{js}\cap E_j} a_{ijt} \;\ge\; n_{js} && \forall j,\ s:\ \pi_{js}\ge 1,\ t\in T_j && (13)\\
& \sum_{i:\, lev_i\ge\lambda_j} x_{ij,\mathrm{lead},t} \;\ge\; 1 && \forall j,\ t\in T_j && (14)\\
& h_{ijt} \;\le\; cap_{it}\, a_{ijt} && \forall i,\ j,\ t\in T_j && (15)\\
& h_{ijt} \;\ge\; \underline{h}\, a_{ijt} && \forall i,\ j,\ t\in T_j && (16)\\
& \sum_{j:\, t\in T_j} h_{ijt} \;\le\; cap_{it} && \forall i,\ t && (17)\\
& \sum_{i\in E_j} h_{ijt} \;\ge\; e_{jt} && \forall j,\ t\in T_j && (18)\\
& \sum_{j:\, t\in T_j} a_{ijt} \;\le\; \overline{n}_i && \forall i,\ t && (19)\\
& \sum_{t\in T_j}\sum_{i\in E_j} c_i\, h_{ijt} \;\le\; B_j && \forall j && (20)\\
& x_{ijrt},\, a_{ijt},\, z_{ij}\in\{0,1\},\quad h_{ijt}\ge 0.
\end{aligned}
$$

Reading the model. (10) says a person active on a project in a period holds exactly one role there. (12) is per-period role coverage (δ_{jr} can be made time-dependent δ_{jrt} if a project needs a reviewer only near delivery). (13) re-imposes skill coverage *in every active period*, so coverage cannot lapse when a team rotates. (14) guarantees the lead in each period is sufficiently senior. (15) links hours to assignment with the *tight* big-M cap_{it} (never use a generic large M here); (16) imposes a minimum meaningful allocation $\underline{h}$ (e.g. 4 hours/week) to forbid sliver assignments. The term $-\mu\sum z_{ij}$ in (9) charges a small fixed cost per team membership, discouraging fragmented teams; μ also controls the classic trade-off between a few heavily-loaded generalists and many lightly-loaded specialists.

4.3 Continuity and churn control

Re-solving a time-indexed model can churn assignments. Two standard, composable devices:

(a) **Team stability within the horizon.** Penalize onboarding events with $u_{ijt}\in\{0,1\}$ ($u\ge a_{ijt}-a_{ij,t-1}$) and add $-\theta\sum u_{ijt}$ to the objective: each time a person *joins* a project mid-stream costs θ.

(b) **Stability across re-solves.** Given the incumbent plan $\hat{a}_{ijt}$, penalize deviations $\sum_{i,j,t}\tau_{ijt}\,|a_{ijt}-\hat{a}_{ijt}|$, linearized in the usual way; or hard-fix assignments inside a frozen window $t\le t_{\mathrm{freeze}}$.

Remark 4.1 (When Variant 2 is worth its size). The variable count is $O(|I|\,|J|\,|R|\,|T|)$. Use it when effort profiles e_{jt} genuinely vary, when part-time multi-project allocation is the norm (matrix organizations, consultancies), or when role hand-offs over time are part of the decision. If everyone works one project at a time full-time, Variant 1 plus a start-date heuristic is usually better value.

5 Variant 3: Soft Coverage, Project Selection, and Penalties

Variants 1–2 presume every project must be staffed and can be. In practice demand exceeds supply: the model should *choose* which optional projects to run, fully staff the mandatory ones, and degrade gracefully — reporting *where* coverage falls short rather than returning INFEASIBLE.

5.1 Additional decision variables

$w_j \in \{0,1\}$	project j is selected (staffed at all),
$u_{js} \geq 0$	shortfall in coverage of skill s on project j,
$q_{jt} \geq 0$	shortfall in effort on project j in period t (time-indexed setting).

5.2 Formulation (built on Variant 1; the same surgery applies to Variant 2)

$$\text{maximize} \quad \sum_j v_j\, w_j \;+\; \alpha \sum_{i,j} f_{ij}\, x_{ij} \;+\; \beta \sum_{i,j} p_{ij}\, x_{ij} \;-\; \gamma \sum_{i,j} c_{ij}\, x_{ij} \;-\; \sum_{j,s} \phi_{js}\, u_{js} \tag{21}$$

$$\text{s.t.} \quad \sum_{i \in Q_{js} \cap E_j} x_{ij} \;\geq\; n_{js}\, w_j \;-\; u_{js} \qquad \forall j,\ s:\ \pi_{js} \geq 1 \tag{22}$$

$$u_{js} \;\leq\; \overline{u}_{js} \qquad \forall j,\ s \tag{23}$$

$$x_{ij} \;\leq\; w_j \qquad \forall j,\ i \in E_j \tag{24}$$

$$\underline{N}_j\, w_j \;\leq\; \sum_{i \in E_j} x_{ij} \;\leq\; \overline{N}_j\, w_j \qquad \forall j \tag{25}$$

$$w_j \;=\; 1 \qquad \forall j \in J^{\text{man}} \tag{26}$$

$$(4),\ (5), \quad x_{ij},\, w_j \in \{0,1\}, \quad u_{js} \geq 0.$$

Design notes.

- **Selection.** An unselected project ($w_j = 0$) absorbs no people, by (24) and (25), and forfeits its value v_j. The model thus performs portfolio triage and staffing *jointly*, which dominates the common two-phase practice of first picking projects and then discovering they cannot be staffed.
- **Soft coverage.** The slack u_{js} converts a hard coverage row into a priced one. Penalties should be *tiered*: $\phi_{js} \gg \max_j v_j$ for mission-critical skills (effectively hard), moderate for nice-to-have depth. The cap (23) prevents the absurd solution of "staffing" a selected project entirely with slack — e.g. $\overline{u}_{js} = n_{js} - 1$ forces at least one qualified person on every required skill of a selected project.

- **Mandatory projects** keep hard coverage simply by fixing $\overline{u}_{js} = 0$ for $j \in J^{\text{man}}$.
- **Diagnosis.** At optimum, the positive u_{js} are exactly the hiring/training gaps: read them as a ranked shopping list for recruitment, with shadow value ϕ_{js} per unit.

In the time-indexed setting the same treatment applies to effort: replace (18) by $\sum_i h_{ijt} \geq e_{jt}\, w_j - q_{jt}$ with penalty $\sum_{j,t} \psi_j\, q_{jt}$, capturing schedule slippage rather than skill gaps.

6 Cross-Cutting Constraint Modules

The following modules attach to any variant. Throughout, read x_{ij} as z_{ij} in the time-indexed setting.

6.1 Pairwise compatibility

$$x_{ij} + x_{i'j} \;\leq\; 1 \qquad \forall (i,i') \in P^-,\ \forall j \tag{27}$$

$$x_{ij} \;=\; x_{i'j} \qquad \forall (i,i') \in P^+,\ \forall j. \tag{28}$$

Constraint (27) covers interpersonal conflicts and manager–report separation (instantiate P^- from the org chart when policy forbids a manager reviewing their own report's project work). The must-pair constraint (28) is strong (both or neither); the weaker "if i then i'" is $x_{ij} \leq x_{i'j}$, useful for apprentice–mentor pairings.

6.2 Synergy bonuses (linearized products)

To *reward* (not require) putting a productive pair together, add $w_{ii'j} \in [0,1]$ with the McCormick equalities for a product of binaries:

$$w_{ii'j} \leq x_{ij}, \qquad w_{ii'j} \leq x_{i'j}, \qquad w_{ii'j} \geq x_{ij} + x_{i'j} - 1, \tag{29}$$

and the objective term $+\sum \sigma_{ii'}\, w_{ii'j}$. With $\sigma_{ii'} > 0$ the first two inequalities suffice (the maximization pushes w up); include the third only for negative synergies. Instantiate these variables *sparsely* — only for pairs with $|\sigma_{ii'}|$ above a threshold — or the model inflates quadratically.

6.3 Team composition and diversity

A ratio cap "no more than a fraction ρ_d of any team from department d" looks nonlinear (team size is a variable) but linearizes exactly:

$$\sum_{i \in I_d \cap E_j} x_{ij} \;\leq\; \rho_d \sum_{i \in E_j} x_{ij} \qquad \forall j,\ d. \tag{30}$$

Floors ($\geq$) work symmetrically, e.g. representation targets across departments or sites. A minimum seniority *mix* beyond the lead rule is the same pattern with I_d replaced by $\{i : lev_i \geq l\}$.

6.4 Location and time-zone overlap

Discretize into zone clusters Z, let I_z be people in cluster z, and require a quorum in a core cluster z_j^* chosen per project:

$$\sum_{i \in I_{z_j^*} \cap E_j} x_{ij} \;\geq\; \eta_j \sum_{i \in E_j} x_{ij}, \tag{31}$$

i.e. at least a fraction η_j of the team shares core working hours. Hard on-site requirements are handled upstream via the eligibility sets E_j (Remark 2.1), not as constraints.

6.5 Fairness of load

Unconstrained cost minimization concentrates work on the cheapest capable people. Two remedies:

(a) **Min–max utilization.** With utilization $util_i = \sum_{j,t} h_{ijt} \big/ \sum_t cap_{it}$, add $\Lambda \geq util_i \;\forall i$ and the term $-\omega\Lambda$ to the objective.

(b) **Balanced project counts.** Bound the spread: $\underline{\nu} \leq \sum_j x_{ij} \leq \overline{\nu}$ for all i in a designated pool (everyone gets between $\underline{\nu}$ and $\overline{\nu}$ assignments).

6.6 Development goals (stretch assignments)

Growth is exposure to a desired skill *without* being counted on for coverage. Let $G_{js} \subseteq I$ be people with $g_{is} = 1$ who are *near*-qualified ($prof_{is} = \pi_{js} - 1$). Introduce $y^g_{ijs} \in \{0,1\}$ (a stretch slot) with

$$y^g_{ijs} \;\leq\; x_{ij}, \qquad \sum_s y^g_{ijs} \;\leq\; 1, \qquad \sum_{i \in G_{js}} y^g_{ijs} \;\leq\; 1 \quad \forall j, s, \tag{32}$$

and the objective bonus $+\epsilon \sum y^g_{ijs}$. Crucially, y^g does *not* appear in the coverage constraints: the stretch person shadows a qualified one (who is still required by (2)). If desired, couple them: $y^g_{ijs} \leq \sum_{i' \in Q_{js}} y_{i'js}$ using the explicit coverage variables of Section 3.3, so a stretch slot exists only where a qualified owner does.

7 Objective Design and Multi-Objective Trade-offs

The criteria in play — skill fit, preference satisfaction, cost, strategic value covered, plus soft penalties — are incommensurable. The formulations above use a **weighted sum**

$$\text{maximize} \quad \alpha\,\text{Fit} + \beta\,\text{Pref} + \sum_j v_j w_j - \gamma\,\text{Cost} - \sum \phi\, u - \sum \psi\, q - \dots \tag{33}$$

which is the practical default, but the weights deserve care.

7.1 Normalization

Raw scales differ wildly (fit scores in $[0,1]$, costs in the 10^5–10^6 range), so raw weights are meaningless. Normalize each term by its *ideal value*: solve the single-objective problem $\max \text{Fit}$ to get Fit^*, likewise $\text{Cost}^* = \min \text{Cost}$ over feasible staffings, then optimize

$$\alpha\,\frac{\text{Fit}}{\text{Fit}^*} - \gamma\,\frac{\text{Cost}}{\text{Cost}^*} + \dots \tag{34}$$

so that α, γ express genuine relative importance. Each ideal value costs one auxiliary MILP solve — cheap, and reusable across weight settings.

7.2 Lexicographic (preemptive) ordering

When priorities are strict — e.g. (1) staff all mandatory projects with zero coverage shortfall, (2) maximize strategic value of optional projects, (3) minimize cost, (4) maximize preferences — solve a sequence of MILPs: optimize objective 1; add the constraint $\mathrm{obj}_1 \geq (1-\varepsilon)\,\mathrm{obj}_1^*$; optimize objective 2 over the restricted set; and so on. The tolerance ε (e.g. 1%) trades a sliver of a higher priority for substantial freedom in lower ones and markedly better solutions in practice than $\varepsilon = 0$.

7.3 ε-constraint method

Keep one objective, bound the rest:

$$\text{maximize Fit} \quad \text{s.t.} \quad \text{Cost} \leq \overline{C}, \qquad \sum_j v_j w_j \geq \underline{V}, \tag{35}$$

then sweep $\overline{C}$ (and/or $\underline{V}$) to trace the Pareto frontier. This is the method of choice for *presenting* the trade-off to decision-makers: "every additional \$50k of staffing budget buys this much fit" is a curve, not a weight.

7.4 Goal programming

When stakeholders state targets ("average utilization near 80%," "preference score at least 0.7 per person"), attach deviation variables to each goal, $\mathrm{goal}_k - d_k^+ + d_k^- = \mathrm{target}_k$, and minimize the weighted sum of the undesirable deviations. This subsumes the soft-coverage device of Variant 3, and it is the honest formulation when management thinks in targets rather than trade-off rates.

7.5 Which to use

Weighted sum for routine automated runs (fast, single solve); lexicographic when governance imposes a strict priority order; ε-constraint for stakeholder-facing what-if analysis; goal programming when the input arrives as targets. They can be mixed: lexicographic at the top (mandatory coverage first), weighted sum within the final level.

8 Computational Considerations

- **Scale.** Variant 1 has $O(|I|\,|J|)$ binaries — typically thousands, easy for modern solvers. Variant 2 has $O(|I|\,|J|\,|R|\,|T|)$; at 200 people $\times$ 40 projects $\times$ 3 roles $\times$ 26 weeks this is $\sim 6 \times 10^5$ binaries, demanding but tractable *if* eligibility preprocessing (Remark 2.1) prunes the majority of (i, j) pairs, as it usually does.

- **Tight big-Ms.** The only big-M in the family is (15), and cap_{it} is its tightest possible value. Resist the generic $M = 10^6$; the LP relaxation gap it opens is the difference between minutes and hours of solve time.

- **Symmetry.** Interchangeable people (identical skills, cost, availability) create symmetric solutions that bloat the branch-and-bound tree. Either aggregate them into pools with integer "count" variables, or add lexicographic ordering cuts $x_{ij} \geq x_{i'j}$-style over each symmetry class. Role *slots* should never be modeled individually ("contributor slot 1, slot 2") — the headcount form (12) is symmetry-free by construction.

- **Rolling horizon.** For long horizons, solve Variant 2 over a window (e.g. 8 weeks detailed + remainder aggregated to Variant-1 granularity), commit the first weeks, roll forward. Combine with the re-solve stability penalty of Section 4.3.
- **Warm starts.** The incumbent staffing plan is almost always feasible for the re-solve after minor data changes; pass it as a MIP start.
- **Infeasibility handling.** Prefer Variant 3's priced slacks to post-hoc IIS analysis: the optimal slack pattern is itself the diagnosis, in the units managers care about.

9 Choosing a Variant: Summary

Variant	Decision granularity	Use when
1. Static, hard coverage	x_{ij}	Portfolio-level team composition; stable long-lived projects; staffing is feasible and mandatory.
2. Time- and role-indexed	x_{ijrt}, h_{ijt}	Varying effort profiles; part-time multi-project work; roles and handoffs matter; capacity is the binding resource.
3. Soft coverage + selection	Variant 1 or 2 + w_j, u_{js}, q_{jt}	Demand exceeds supply; portfolio triage is part of the question; infeasibility must degrade into priced, diagnosable shortfalls.

The modules of Section 5 — pairwise compatibility, synergy, diversity ratios, time-zone quorum, fairness, stretch assignments — bolt onto any row of this table, and the objective machinery of Section 6 applies uniformly. In practice the recommended path is: start with Variant 1 plus soft coverage (the Variant-3 surgery), validate the data and weights with ε-constraint sweeps, and move to Variant 2 only for the projects and people where period-level capacity is demonstrably the binding constraint.